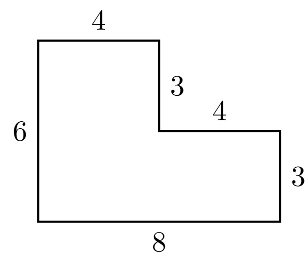
Composite Rectilinear Figures: Area and Perimeter

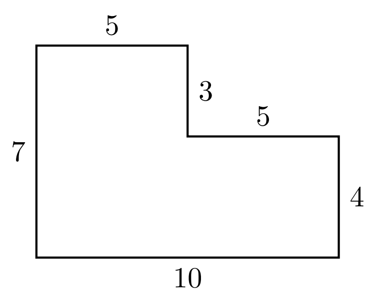
Skills practised: Adding rectangles, subtracting a corner, going around the outside, and finding a missing side. *Every shape here is made only of rectangles, and every corner is a square corner.* Read each question carefully — some ask for **area** (how much space is covered, in square units), some ask for **perimeter** (how far it is all the way around), and some ask for a **missing side length**. Show your work in the space beside each figure.

1. The figure shows a paper sticker, measured in centimetres. Cut it into two rectangles with one straight line, find the area of each rectangle, then add. What is the **area** of the whole sticker?



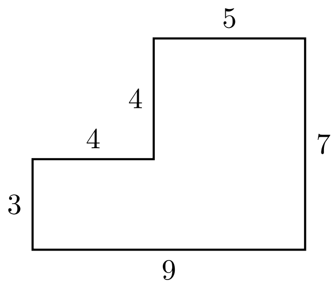
Answer: ________________ cm^2

2. A ribbon is glued all the way around the edge of this shape, measured in centimetres. How long is the ribbon? That is, what is the **perimeter**?



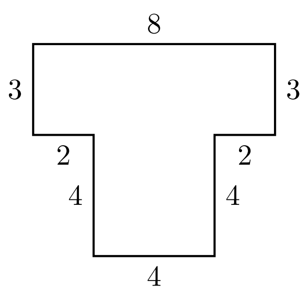
Answer: ________________ cm

3. Find the **area** of this shape by breaking it into two rectangles. (Lengths are in centimetres.)



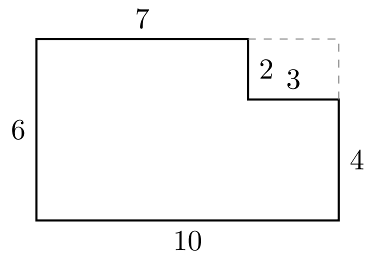
Answer: ________________ cm^2

4. This name tag has eight sides, measured in centimetres. Find its **perimeter** by adding all eight side lengths.



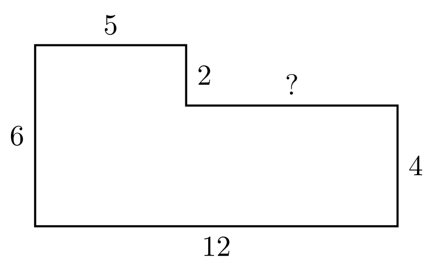
Answer: ________________ cm

5. Maya says: "I can find the area a different way. I start with the whole big rectangle and take away the corner that is missing." Use Maya's way to find the **area** of this shape. (Lengths are in centimetres; the dashed lines show the missing corner.)



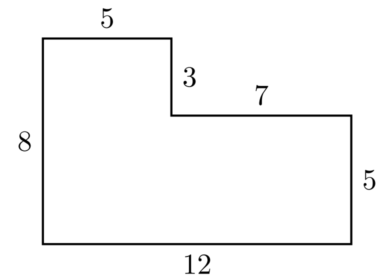
Answer: ________________ cm^2

6. One side length has been rubbed out and marked with a ?. The bottom of the shape is 12 cm long and the top piece is 5 cm long. How long is the side marked ?



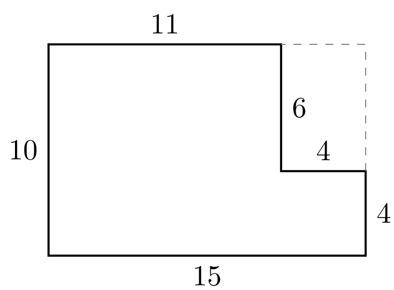
Answer: ________________ cm

7. A reading-corner rug has the shape below, measured in feet. How many square feet of carpet does it cover? (Find the **area**.)



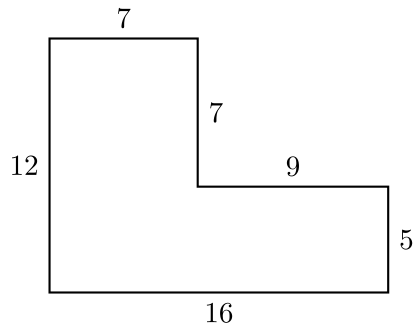
Answer: ________________ ft^2

8. A patio is a 15 ft by 10 ft rectangle with a rectangular flower bed cut out of one corner. The flower bed is 4 ft across and 6 ft deep. What is the **area** of the paved part that is left?



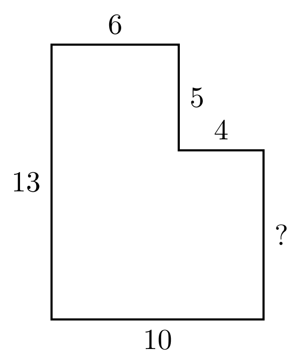
Answer: ________________ ft^2

9. A vegetable garden has the shape below, measured in feet. A fence goes all the way around the outside. How many feet of fencing are needed? (Find the **perimeter**.)



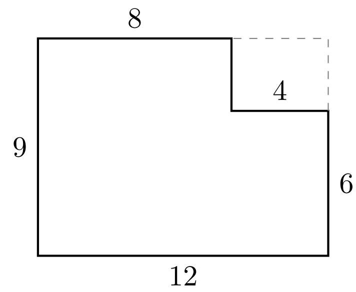
Answer: ________________ ft

10. The whole left side of this deck measures 13 ft, and the upper piece on the right measures 5 ft. How long is the side marked ? Be ready to explain why the two vertical pieces on the right must add up to the long side on the left.



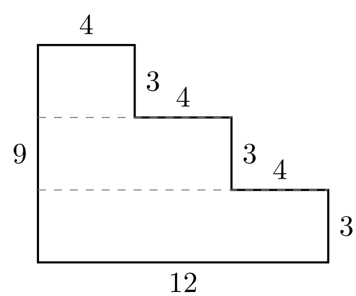
Answer: ________________ ft

11. **Challenge.** One side of this porch floor is *not* labelled. Work out that missing length first, then find the **area** of the porch. (Lengths are in feet.)



Answer: ________________ ft^2

12. **Challenge.** This stage is built like three steps (lengths in feet). Break it into *three* rectangles that each lie flat like a step, find the area of each one, then add them. What is the total **area**?



Answer: ________________ ft^2